Software Documentation

1. Software Overview

The software enables a **gesture-controlled wireless system** where the user's hand movements are captured and translated into real-time motion of a turret. The sender ESP32 reads orientation data from the MPU6050 sensor, processes it into pitch and roll values and transmits this data wirelessly using ESP-NOW. The receiver ESP32 interprets this data to control servo motors for directional movement, while also managing inputs from an ultrasonic sensor to detect obstacles and trigger a laser module for visual feedback.

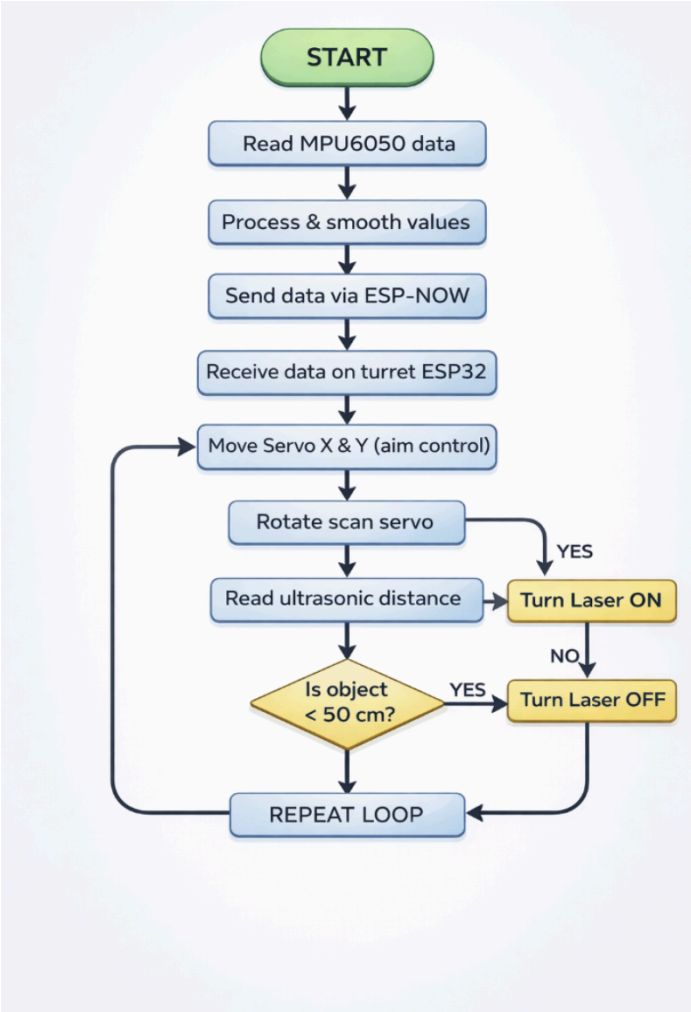
Platforms used:

1. Thonny: for basic python

2. System Logic

1. MPU6050 detects hand tilt (pitch and roll)
2. Sender ESP32 processes and smooths the sensor data
3. Processed data is sent wirelessly using ESP-NOW
4. Receiver ESP32 receives the data
5. Servo motors move accordingly (X and Y axis for aiming)
6. Third servo continuously scans using ultrasonic sensor
7. Ultrasonic sensor measures distance
8. If object is within range (< 50 cm), laser turns ON
9. If no object is detected, laser remains OFF

3. Flowchart



4. App Flow (If MIT App Inventor)

The mit app inventor was not used for this project.